Course Catalog

Complete Research & Content Guide

5 Courses · A to Z Content · 90-Day Roadmaps · Verified Resources

Course	Target Profile	Time to Job-Ready	Fresher Salary (India)
1. GenAI & Prompt Engineering	Anyone curious about AI, any background	60-90 days	INR 6-12 LPA
2. DevOps & Cloud Engineering	Tech-minded, problem-solvers, sysadmin interest	90 days	INR 6-14 LPA
3. Frontend Development	Visual thinkers, creators, builders	90 days	INR 4-8 LPA
4. Data Analytics	Analytical, numbers-driven, business-curious	90 days	INR 4-7 LPA
5. Cybersecurity & Ethical Hacking	Detail-oriented, security-curious, patient	90 days	INR 4-8 LPA

How to Use This Document

This document is the single source of truth for all course content in the platform.

Each course section contains: Overview, Who It's For, Skills Learned, 90-Day Week-by-Week Roadmap,

Free Resources, Paid Resources, Projects to Build, Career Outcomes, and Job Titles.

All salary data is India-specific and sourced from 2025-2026 market research.

All resource links are verified as of April 2026.

COURSE 1 OF 5

GenAI & Prompt Engineering

One-Line Description

Learn to harness the power of Generative AI — from crafting perfect prompts to building real AI-powered applications using Python, LangChain, and modern LLM APIs.

Course Overview

Field	Detail
Full Name	Generative AI & Prompt Engineering
Difficulty	Beginner to Intermediate — No prior coding needed to start
Prerequisites	Basic computer literacy. Python knowledge is a bonus, not required.
Duration	90 Days (2-3 hours/day)
Time to Income	60-90 days — freelance automation gigs possible in Month 2
Best For	Students, non-tech graduates, marketers, anyone who uses ChatGPT daily
NOT For	People who hate writing or communicating clearly; this is a language-heavy field
Salary (Fresher)	INR 6-12 LPA (Junior Prompt Engineer, AI Analyst, AI Automation Executive)
Salary (Mid)	INR 12-25 LPA (LLM Engineer, GenAI Developer, AI Product Manager)
Salary (Senior)	INR 25-45+ LPA (Senior GenAI Engineer, AI Architect)
Top Hiring Companies	TCS, Infosys, Wipro, Razorpay, Freshworks, startups, global remote

What This Field Actually Is

Generative AI refers to AI systems that create new content — text, images, code, audio — by learning patterns from massive datasets. When you type a message into ChatGPT and it writes you a full blog post, that is generative AI in action.

Prompt engineering is the art and science of writing instructions to AI models that produce the exact output you want. A poor prompt gives a generic answer. A well-engineered prompt gives you a precise, structured, high-quality result — every time.

By the end of this course, you will be able to: build chatbots, automate business workflows with AI, integrate LLM APIs into web apps, use RAG (Retrieval-Augmented Generation) to make AI answer questions from your own documents, and create AI agents that can browse the web, read files, and take actions autonomously.

Who This Course Is For

- Non-tech graduates who want to enter the AI field without a CS degree
- Marketing or content professionals who want to automate their work using AI
- Developers who want to add AI capabilities to their existing projects
- Entrepreneurs who want to build AI-powered products quickly
- Anyone who already uses ChatGPT daily and wants to go deeper

Who This Course Is NOT For

- People who want to build AI models from scratch — this is about using models, not building them
- Those who hate reading documentation — this field requires constant curiosity
- People looking for a fully passive income — this requires active skill-building

Skills You Will Have After 90 Days

Skill	Level After Course	Real-World Use
Prompt Engineering (Zero-shot, Few-shot, Chain-of-Thought)	Intermediate	Write prompts that get precise, reliable AI outputs every time
Python for AI	Beginner-Intermediate	Write scripts to automate tasks with AI APIs
OpenAI / Gemini API Integration	Intermediate	Connect AI models to apps, send requests, parse responses
LangChain Basics	Beginner	Build multi-step AI workflows and chatbots with memory
RAG (Retrieval-Augmented Gen)	Beginner	Make AI answer questions from your own PDFs and documents
AI Agents	Beginner	Build agents that can search the web, read files, take action
Vector Databases	Beginner	Store and search document embeddings (ChromaDB, Pinecone)
Prompt Templates & Libraries	Intermediate	Maintain a reusable prompt library for consistent outputs
AI Automation Workflows	Intermediate	Automate content creation, email drafting, data extraction
Multi-modal Prompting	Beginner	Prompt AI with text + images using GPT-4o, Gemini Vision

90-Day Week-by-Week Roadmap

Phase 1: Foundation

Understand how AI works, master prompt engineering, write your first Python scripts

Week 1: AI Fundamentals & Tool Setup

Daily commitment: 1.5-2 hours/day

- Understand what LLMs are: how GPT, Gemini, and Claude actually work (transformers, tokens, temperature)
- Create accounts: OpenAI, Google AI Studio (Gemini), Hugging Face, Anthropic
- Explore ChatGPT, Claude, and Gemini — side-by-side comparison on same prompts
- Learn basic prompt anatomy: Role + Task + Context + Format + Constraints
- Write 20 prompts for different scenarios: email, code, analysis, creative writing
- Setup VS Code and install Python 3.11+ — follow official docs step by step

Milestone: You can explain what an LLM is in plain English and write structured prompts that consistently produce good outputs

Week 2: Core Prompting Techniques

Daily commitment: 2 hours/day

- Zero-shot prompting: task with no examples — practice 10 examples
- Few-shot prompting: give 2-3 examples before your task — practice with classification tasks
- Chain-of-Thought prompting: ask the AI to reason step-by-step before answering — great for math and logic
- Role-based prompting: 'You are a senior software engineer at Google...'
- Multi-turn prompting: refine answers across a conversation
- Create your first personal prompt library in Notion — organized by category (writing, coding, analysis, research)
- Practice: Write a prompt that generates a complete Indian startup pitch deck outline

Milestone: You have a prompt library of 30+ templates and can reliably get high-quality outputs for 5+ task types

Week 3: Python for AI — Absolute Basics

Daily commitment: 2-2.5 hours/day

- Install Python 3.11, pip, and set up a virtual environment
- Python basics: variables, strings, lists, dictionaries, if/else, for loops, functions
- Learn f-strings: essential for building dynamic prompts programmatically
- Install and use the openai Python library: pip install openai
- Write your first API call to ChatGPT in Python — 15 lines of code
- Build a simple command-line tool: type a prompt, get a response, save to a file
- Read the OpenAI documentation — learn about max_tokens, temperature, system messages

Milestone: You can write a Python script that calls the ChatGPT API, sends a dynamic prompt, and saves the output

Week 4: Advanced Prompting & Automation

Daily commitment: 2-2.5 hours/day

- Structured output prompting: instruct AI to respond in JSON format — parse with Python
- System message engineering: define AI persona and behavior at the session level
- Iterative refinement: build a loop that keeps refining the AI output until it meets criteria
- Build Prompt v1.0: an automated email writer — takes bullet points, outputs professional email
- Build Prompt v2.0: a social media content generator for 3 platforms (LinkedIn, Twitter/X, Instagram)
- Error handling in Python: what to do when the API fails or returns unexpected output
- Introduction to the Gemini API — repeat the same projects using Google's API

Milestone: You can build a complete AI automation tool using Python that accepts user input and produces structured AI output

Phase 2: Hands-On Building

Build real AI applications — chatbots, RAG systems, and AI agents

Week 5: LangChain Fundamentals

Daily commitment: 2.5 hours/day

- Install LangChain: pip install langchain langchain-openai
- Understand LangChain's core components: LLMs, Chains, Prompts, Memory, Agents, Tools
- Build your first LangChain chain: a simple Q&A bot with a custom system prompt
- Add memory to your chain: the AI now remembers previous messages in the conversation
- LangChain prompt templates: create dynamic prompts with variables
- Build a simple customer support chatbot using LangChain + OpenAI
- Deploy it as a command-line tool — test with 10 different user scenarios

Milestone: You have a working chatbot built with LangChain that maintains conversation history

Week 6: RAG — Make AI Answer from Your Documents

Daily commitment: 2.5-3 hours/day

- Understand what RAG is: AI searches your documents first, then answers using that context
- Install: pip install langchain chromadb pypdf sentence-transformers
- Load a PDF in Python, split into chunks using RecursiveCharacterTextSplitter
- Create embeddings using OpenAI Embeddings API and store in ChromaDB
- Build a similarity search: given a question, find the most relevant document chunks
- Connect RAG to LangChain: question → retrieve relevant chunks → send to LLM → get answer

- Project: Build a PDF Q&A bot — upload any PDF, ask questions, get accurate answers with source citations

Milestone: You can build a RAG system that answers questions accurately from any uploaded document

Week 7: AI Agents — AI That Takes Action

Daily commitment: 3 hours/day

- Understand AI agents: LLM + tools + memory + a goal = an agent that can complete tasks autonomously
- Install LangChain agents toolkit and learn about the ReAct framework
- Give your agent tools: web search (via SerpAPI or Tavily), calculator, Python REPL
- Build Agent v1: a research agent — give it a topic, it searches the web, reads articles, writes a summary
- Understand tool calling in OpenAI API (function calling) — define tools as JSON schemas
- Build Agent v2: a data extraction agent — give it a website URL, it extracts structured data
- Introduction to multi-agent systems: two agents working together on a task

Milestone: You have a working AI agent that can autonomously search the web and synthesize information

Week 8: Capstone Project — Build an AI Product

Daily commitment: 3 hours/day

- Choose one of these capstone projects: (1) AI-powered business assistant for a specific industry, (2) YouTube video summarizer that extracts key points from any video, (3) AI homework helper that explains any concept using examples
- Build the full system: frontend (simple HTML form or Streamlit), backend (Python + LangChain), and output display
- Add error handling, loading states, and clean UI
- Write documentation: what it does, how to run it, what APIs it uses
- Deploy on Hugging Face Spaces (free) or Replit — get a live public URL
- Record a 2-minute demo video of your project
- Share on LinkedIn with technical breakdown — this is your first portfolio piece

Milestone: You have a deployed, live AI application that you can demo to potential employers or clients

Phase 3: Job Ready

Build portfolio, apply for jobs, start freelancing

Week 9: Portfolio & Resume

Daily commitment: 2 hours/day

- Organize your 3-4 best projects on GitHub with clean README files for each
- Create a portfolio page on Notion or a simple HTML site — showcase projects with screenshots and live links
- Write your resume: highlight AI tools (LangChain, OpenAI API, Gemini API, RAG, Python), projects, and outcomes

- Optimize LinkedIn: headline should say 'GenAI Developer | LLM Applications | Python'
- Write 3 LinkedIn posts about your projects — explain what you built and what you learned
- Get your projects reviewed by someone in the field — post in AI communities for feedback

Milestone: Your GitHub, LinkedIn, and portfolio all tell a coherent story of your AI skills

Week 10: Job Hunting — Employed Route

Daily commitment: 3+ hours/day

- Apply to 5-10 roles per day on LinkedIn, Naukri, Instahyre, AngelList India
- Search for: Prompt Engineer, Junior AI Developer, AI Analyst, Automation Engineer, LLM Developer
- Tailor your resume for each application — match keywords from the job description
- Cold connect on LinkedIn with 5 recruiters at target companies per day with a personalized note
- Prepare for technical interviews: explain how RAG works, what embeddings are, how agents work
- Mock interviews: use ChatGPT to simulate AI interview questions — practice out loud

Milestone: You have 50+ active applications and are getting interview calls

Week 11: Freelancing Route — First Client

Daily commitment: 3+ hours/day

- Create a profile on Upwork: title 'AI Automation Expert | ChatGPT API | LangChain | Python'
- Write a strong profile description with your portfolio links and project outcomes
- Bid on 5-10 projects per day in these categories: AI Chatbot Development, Prompt Engineering, AI Automation, ChatGPT Integration
- Offer competitive rates to start: INR 1500-3000/hour or fixed price INR 5000-15000 per small project
- Find local businesses that need AI integration: automated customer service, document Q&A, content generation
- Close your first project — even if it is small. Reviews and ratings matter more than money at this stage.

Milestone: You have landed your first paid freelance project or are actively interviewing at companies

Week 12: Specialize & Level Up

Daily commitment: 2 hours/day

- Pick one specialization: (1) AI for content/marketing automation, (2) AI for business intelligence, (3) AI application development
- Learn OpenAI Assistants API — the next evolution beyond basic API calls
- Explore fine-tuning: when and why you would fine-tune a model vs use prompt engineering
- Learn about multi-modal AI: GPT-4o and Gemini can process images — add this to your toolkit

- Follow AI researchers on X (Twitter): Andrej Karpathy, Sam Altman, Harrison Chase (LangChain)
- Join AI communities: r/ChatGPT, r/LangChain, Hugging Face Discord, AI Agents India (LinkedIn group)
- Negotiate or close job offer / increase freelance rate based on first completed project

Milestone: You have a specialization, an active income source (job or freelance), and a clear 6-month growth plan

Free Resources

Resource Name	Type	Cost	Why Use It
Andrej Karpathy — Neural Networks: Zero to Hero	YouTube Series	Free	Best explanation of how LLMs work at a fundamental level — by the former Tesla AI director
Google AI Studio (Gemini)	Playground	Free	Experiment with Gemini models — free tier is generous, great for prompt testing
OpenAI Cookbook	Documentation	Free	Official examples and guides from OpenAI — copy-paste code that actually works
LangChain Documentation	Documentation	Free	Official docs — start with Quickstart, then LCEL (LangChain Expression Language)
Prompt Engineering Guide (promptingguide.ai)	Website	Free	The most comprehensive free guide to every prompting technique — bookmark this
freeCodeCamp — Generative AI Full Course	YouTube	Free	6-hour video covering LLMs, APIs, LangChain, and building AI apps from scratch
Hugging Face Course	Online Course	Free	Official NLP and transformer course — covers the theory behind LLMs
DeepLearning.AI Short Courses	Online Courses	Free	Andrew Ng's 1-2 hour courses on LangChain, RAG, agents, prompt engineering
roadmap.sh/prompt-engineering	Roadmap	Free	Visual roadmap of every concept to learn — good reference for what's next
r/PromptEngineering (Reddit)	Community	Free	Real prompt examples, discussions, job postings — very active community

Paid Resources (Worth the Money)

Resource Name	Type	Cost	Why Use It
DeepLearning.AI — LangChain for LLM App Dev	Course	~INR 3000 (Coursera)	By LangChain's creator Harrison Chase — the definitive course, 1 week to complete
IBM Generative AI Engineer Specialization	Course	~INR 3500/month Coursera	Structured 6-course program, good for getting a formal certificate
Udemy — LangChain: Develop LLM-powered apps	Course	~INR 500 (sale price)	Hands-on coding course — practical, project-based, well-rated
KodeKloud — AI for DevOps	Course	~INR 1500/month	Good if you want to combine AI with cloud/DevOps automation

Projects to Build

Project 1: AI Email Assistant

- What you build: A Python tool that takes bullet points of what you want to say and generates a professional email. Input: bullets. Output: 3 email variations (formal, friendly, brief).
- Skills demonstrated: OpenAI API, Python, prompt templates, output formatting
- Estimated build time: 3-5 hours
- Best done in: Phase 1

Project 2: PDF Question Answering Bot

- What you build: Upload any PDF document (textbook, legal contract, financial report) and ask it questions. The bot finds the relevant section and answers accurately with source citations.
- Skills demonstrated: RAG, ChromaDB, LangChain, PyPDF, embeddings
- Estimated build time: 8-12 hours
- Best done in: Phase 2

Project 3: YouTube Video Summarizer

- What you build: Paste a YouTube URL, get a structured summary: 5 key takeaways, main arguments, and a list of action items. Uses YouTube transcript API + LLM.
- Skills demonstrated: YouTube API, LangChain, summarization chain, structured output
- Estimated build time: 6-10 hours
- Best done in: Phase 2

Project 4: AI Research Agent

- What you build: Give it a topic (e.g., 'Impact of GenAI on Indian IT sector') and it autonomously searches the web, reads 5 articles, and writes a structured 500-word report with sources.
- Skills demonstrated: LangChain agents, Tavily search API, ReAct framework, tool use
- Estimated build time: 10-15 hours
- Best done in: Phase 3

Project 5: AI Customer Support Bot (Capstone)

- What you build: A domain-specific chatbot trained on your own FAQ document. Integrates into a simple Streamlit web app. Users type questions, bot answers from your knowledge base only.
- Skills demonstrated: RAG, LangChain, Streamlit, ChromaDB, conversation memory
- Estimated build time: 15-20 hours
- Best done in: Phase 3

Career Outcomes

Job Title	Experience Needed	Salary Range (India)	What You Do
Prompt Engineer	0-1 year	INR 6-15 LPA	Design and optimize prompts for AI systems at scale
Junior AI Developer	0-1 year	INR 6-12 LPA	Build AI-powered features in web apps using LLM APIs
AI Automation Executive	0-1 year	INR 5-10 LPA	Automate business processes using AI tools and scripts
GenAI Engineer	1-3 years	INR 12-25 LPA	Design and deploy full GenAI systems including RAG and agents
LLM Engineer	2-4 years	INR 20-40 LPA	Fine-tune models, optimize inference, build production LLM pipelines
AI Product Manager	3-5 years	INR 25-60 LPA	Lead AI product strategy and roadmap at tech companies
AI Architect	5+ years	INR 40-80 LPA	Design enterprise-scale AI architectures and systems
Freelance AI Consultant	6+ months	INR 1500-5000/hr	Independent project-based work for businesses adopting AI

Top Companies Hiring GenAI Talent in India

- TCS, Infosys, Wipro — large IT services companies hiring AI automation teams

- Razorpay, Zepto, Swiggy, CRED, Meesho — Indian product companies building AI features
- Freshworks, Zoho — SaaS companies integrating AI into their products
- Global startups via LinkedIn Remote: many US/UK AI startups hire Indian talent remotely
- Freelance platforms: Upwork, Toptal, Contra — strong demand for LangChain and API developers

Certifications Worth Getting

- Google Professional Machine Learning Engineer — free learning path, exam costs ~$200
- IBM AI Engineering Specialization (Coursera) — recognized certificate, ~3 months at part-time
- DeepLearning.AI Specializations — Andrew Ng's courses carry strong brand recognition in India
- AWS Certified Machine Learning Specialty — if you want to combine GenAI with cloud

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COURSE 2 OF 5

DevOps & Cloud Engineering

One-Line Description

Master the tools that power modern software delivery — from Linux and Git to Docker, Kubernetes, CI/CD pipelines, and AWS cloud infrastructure.

Course Overview

Field	Detail
Full Name	DevOps Engineering & Cloud Computing
Difficulty	Intermediate — Basic computer skills required. Coding experience helpful but not mandatory.
Prerequisites	Comfortable with computers, basic understanding of what a server is, patience for debugging
Duration	90 Days (2-3 hours/day)
Time to Income	90 days — entry-level DevOps roles and cloud support positions
Best For	CS/IT graduates, system admins, network engineers pivoting to cloud
NOT For	People who get frustrated easily — DevOps requires patience and methodical debugging
Salary (Fresher)	INR 6-14 LPA (Cloud Support Engineer, Junior DevOps Engineer)
Salary (Mid)	INR 15-25 LPA (DevOps Engineer, SRE, Cloud Engineer)
Salary (Senior)	INR 30-60+ LPA (Senior DevOps, Platform Engineer, Cloud Architect)
Top Companies	Infosys, TCS, Wipro, Amazon AWS, Razorpay, Zomato, startups, MNCs

What This Field Actually Is

DevOps is the practice of combining software development (Dev) and IT operations (Ops) to shorten the development lifecycle and deliver software faster, more reliably, and with fewer failures. A DevOps engineer is the person who makes sure code written by developers reaches users quickly, safely, and automatically.

Cloud computing means running software and storing data on remote servers (AWS, Azure, GCP) instead of your own computer or a physical server in your office. Cloud engineers provision, manage, and optimize these cloud resources.

In practice, a DevOps/Cloud engineer spends their day: automating deployment pipelines so code ships automatically, managing containers (Docker/Kubernetes) that package applications, writing Infrastructure as Code (Terraform) to create cloud resources programmatically, monitoring systems for performance and errors, and ensuring security and scalability.

Who This Course Is For

- CS/IT graduates who want a high-paying alternative to software development
- System administrators and network engineers who want to modernize their skills
- Anyone who wants to understand how Netflix, Swiggy, or Zomato's infrastructure works
- People who like problem-solving, debugging, and system design
- Those interested in cloud certifications (AWS, Azure) for job security

Skills You Will Have After 90 Days

Skill	Level After Course	Real-World Use
Linux Command Line	Intermediate	Navigate servers, manage files, run scripts, troubleshoot issues
Git & GitHub	Intermediate	Version control, branching strategies, pull requests, code reviews
Bash Scripting	Beginner-Mid	Automate repetitive server tasks with shell scripts
Docker	Intermediate	Containerize applications, write Dockerfiles, manage images and containers
Docker Compose	Beginner	Run multi-container apps locally with a single YAML file
Kubernetes (K8s)	Beginner	Deploy and manage containerized apps in a cluster
CI/CD (GitHub Actions)	Intermediate	Automate build, test, and deploy pipelines triggered by code pushes
AWS (EC2, S3, IAM, VPC)	Beginner-Mid	Deploy apps on AWS, manage cloud resources, understand cloud networking
Terraform (IaC)	Beginner	Define and provision cloud infrastructure using code — no manual clicking
Monitoring (Prometheus/Grafana)	Beginner	Set up dashboards and alerts to monitor live application performance
Python Scripting	Beginner	Write automation scripts for DevOps tasks

90-Day Week-by-Week Roadmap

Phase 1: Foundation

Linux, Git, networking — the bedrock everything else is built on

Week 1: Linux Fundamentals

Daily commitment: 2 hours/day

- Install Ubuntu 22.04 on VirtualBox (free) or use Ubuntu via WSL2 on Windows / native on Mac
- Navigate the filesystem: ls, cd, pwd, mkdir, rm, cp, mv — practice 50 commands
- File permissions: chmod, chown — understand read/write/execute for user/group/others
- Process management: ps, top, htop, kill, systemctl — start/stop services
- Package management: apt install, apt update, apt upgrade
- Text editing with vim: modes, saving, quitting — this is essential for server work
- Resource: Linux Journey (linuxjourney.com) — free, interactive, browser-based

Milestone: You can confidently navigate a Linux server via command line and manage files, permissions, and processes

Week 2: Networking & Git

Daily commitment: 2 hours/day

- Networking basics: IP addresses, DNS, HTTP/HTTPS, TCP/UDP, ports — how data moves on the internet
- SSH: connect to a remote server, generate SSH keys, configure authorized_keys
- Git fundamentals: git init, clone, add, commit, push, pull — the daily workflow
- Branching: git branch, checkout, merge — the core collaboration pattern
- GitHub: create repositories, open pull requests, review code, resolve merge conflicts
- Practice: Set up a VProfile project — spin up multiple VMs with Vagrant, configure services, deploy manually
- Resource: Pro Git Book (git-scm.com/book) — free, the definitive Git reference

Milestone: You can SSH into a remote server, manage files via command line, and collaborate on code using Git and GitHub

Week 3: Bash Scripting & Python

Daily commitment: 2 hours/day

- Bash variables, conditionals (if/else), loops (for/while), functions
- Write scripts for common DevOps tasks: automated backups, log rotation, health checks
- crontab: schedule scripts to run automatically at set times
- Python basics for DevOps: read/write files, make HTTP requests, parse JSON
- Python script: monitor a website's status every 5 minutes and send an alert if it goes down
- Practice: Write a Bash script that automatically provisions a new Ubuntu server with your standard tools
- Read: The Linux Command Line by William Shotts (free PDF at linuxcommand.org)

Milestone: You can write Bash scripts that automate repetitive server tasks and schedule them with cron

Week 4: CI/CD with GitHub Actions

Daily commitment: 2.5 hours/day

- Understand CI/CD: what it means, why it matters, how it saves hours of manual work
- GitHub Actions: workflows, triggers, jobs, steps, runners
- Build your first pipeline: on every push to main, automatically run tests, build the app
- Add deployment step: automatically deploy to a test server after successful build
- Artifacts and caching: speed up pipelines by caching dependencies
- Secrets management: store API keys and passwords in GitHub Secrets (never in code)
- Project: Build a CI/CD pipeline for a simple Python web app — push code → tests run → app deploys automatically

Milestone: You have a working CI/CD pipeline on GitHub Actions that automatically tests and deploys code on every push

Phase 2: Containers & Cloud

Docker, Kubernetes, and AWS — the core DevOps toolset

Week 5: Docker Fundamentals

Daily commitment: 2.5 hours/day

- What is containerization and why it solved the 'works on my machine' problem
- Install Docker Desktop — understand images vs containers
- Write your first Dockerfile: FROM, RUN, COPY, EXPOSE, CMD
- Build and run Docker images: docker build, docker run, docker ps, docker logs
- Docker networking: bridge networks, port mapping (-p 8080:80)
- Docker volumes: persist data outside containers
- Docker Compose: define multi-container apps (web + database) in a single docker-compose.yml
- Project: Containerize a Node.js or Python Flask web app and run it with Docker
- Resource: Docker's official 'Get Started' guide — free, the best beginner tutorial

Milestone: You can containerize any web application using Docker and run it consistently across any machine

Week 6: Kubernetes (K8s) Basics

Daily commitment: 3 hours/day

- Why Kubernetes: Docker runs one container, K8s orchestrates thousands across many servers
- K8s architecture: Master Node, Worker Nodes, Pods, ReplicaSets, Deployments, Services, Ingress
- Install minikube locally — run a single-node K8s cluster on your laptop
- kubectl commands: apply, get, describe, logs, delete, port-forward
- Deploy your Docker app to K8s: write a Deployment YAML and a Service YAML

- Scaling: kubectl scale deployment my-app --replicas=5 — instantly run 5 copies of your app
- ConfigMaps and Secrets: manage configuration and sensitive data in K8s
- Resource: KodeKloud free Kubernetes course — browser-based labs, no setup required

Milestone: You can deploy a containerized application to a Kubernetes cluster and scale it up and down on demand

Week 7: AWS Cloud — Core Services

Daily commitment: 3 hours/day

- Create an AWS account — use the Free Tier (valid for 12 months, generous limits)
- IAM: users, groups, roles, policies — the security foundation of AWS. Never use root account.
- EC2: launch a virtual machine on AWS, SSH into it, deploy your app
- S3: create buckets, upload files, configure static website hosting, set bucket policies
- VPC: subnets, route tables, internet gateways, security groups — your private network in the cloud
- Elastic Load Balancer: distribute traffic across multiple EC2 instances
- RDS: managed database service — launch a PostgreSQL instance
- Resource: AWS Free Tier (aws.amazon.com/free) — use real AWS services for free
- Resource: AWS Skill Builder (free tier available) — official AWS learning platform

Milestone: You can deploy a full web application stack on AWS using EC2, S3, RDS, and a load balancer

Week 8: Terraform — Infrastructure as Code

Daily commitment: 3 hours/day

- What is IaC: define infrastructure in code files instead of clicking in the console
- Install Terraform, configure AWS credentials
- Terraform basics: providers, resources, variables, outputs, state file
- Write Terraform code to provision: an EC2 instance, an S3 bucket, a security group
- terraform init, plan, apply, destroy — the core workflow
- Modules: reusable Terraform configurations
- Remote state: store Terraform state in S3 so teams can collaborate
- Project: Provision a complete 3-tier application infrastructure on AWS using Terraform — VPC, EC2 instances, RDS, load balancer — all with code

Milestone: You can provision an entire AWS environment using Terraform code — no manual clicking in the console

Phase 3: Monitoring & Job Ready

Observability, security, portfolio, and job applications

Week 9: Monitoring & Observability

Daily commitment: 2.5 hours/day

- Why monitoring matters: you cannot fix what you cannot see

- Prometheus: pull-based metrics collection — install and configure on your K8s cluster
- Grafana: connect to Prometheus, build dashboards for CPU, memory, request rate, error rate
- Alerting: configure alerts in Grafana — get notified when error rate exceeds 5%
- Logging: understand the ELK Stack concept (Elasticsearch, Logstash, Kibana) — set up basic logging
- AWS CloudWatch: monitor EC2 instances, set alarms, view logs — free with AWS account
- Project: Set up a complete monitoring stack for your deployed application — Prometheus + Grafana dashboard + alerts

Milestone: You have a live monitoring dashboard showing real metrics for your deployed application

Week 10: Security & DevSecOps Basics

Daily commitment: 2 hours/day

- Security fundamentals for DevOps: principle of least privilege, secret management, vulnerability scanning
- AWS IAM best practices: MFA, role-based access, no root API keys
- Docker security: scan images with Trivy (free), never run containers as root, use non-root users in Dockerfiles
- GitHub Actions security: use encrypted secrets, pin action versions, limit permissions
- HashiCorp Vault basics: proper secret management for production environments
- OWASP Top 10 for DevOps: the 10 most critical security risks to understand
- Practice: Audit your existing projects for security issues — fix at least 5 things you find

Milestone: You understand DevSecOps principles and can implement basic security controls in your CI/CD pipelines

Week 11: Capstone Project & Portfolio

Daily commitment: 3+ hours/day

- Build the capstone project: a complete DevOps pipeline for a web application — GitHub code → GitHub Actions CI/CD → Docker container → K8s deployment on AWS EKS → Prometheus + Grafana monitoring → Terraform for infrastructure
- Document everything in a detailed GitHub README with architecture diagrams (use draw.io free)
- Write a LinkedIn post explaining your project architecture — this gets recruiter attention
- Prepare your resume: list all tools (Linux, Git, Docker, K8s, GitHub Actions, AWS, Terraform, Prometheus)
- Start applying: LinkedIn, Naukri, Instahyre, AngelList India — search 'DevOps Engineer', 'Cloud Engineer', 'SRE'
- Reach out to 5 DevOps engineers on LinkedIn per day — ask for 15-minute informational calls

Milestone: Your capstone project is live and documented, and you have sent 50+ job applications

Week 12: Interview Prep & Certification

Daily commitment: 2-3 hours/day

- Common DevOps interview questions: explain how Docker works, what is the difference between a container and a VM, how does K8s handle pod failures, walk me through a CI/CD pipeline you built
- System design basics: how would you design the infrastructure for an app that needs to handle 1 million users?
- Live debugging exercises: 'the deployment is failing, walk me through your debugging process'
- Start AWS Cloud Practitioner or AWS Solutions Architect Associate exam prep (free study resources on AWS Skill Builder)
- Mock interviews: use Pramp for system design, LeetCode for any coding rounds
- Negotiate confidently: entry-level DevOps roles in India pay INR 6-14 LPA — know your worth

Milestone: You are interview-ready for junior DevOps and cloud engineering roles

Free Resources

Resource Name	Type	Cost	Why Use It
TechWorld with Nana (YouTube)	YouTube Channel	Free	Best YouTube channel for DevOps — Nana explains Docker, K8s, CI/CD better than any course
Linux Journey (linuxjourney.com)	Interactive Site	Free	Browser-based Linux learning with exercises — perfect for absolute beginners
KodeKloud Free Courses	Online Courses	Free	Docker for Beginners, Kubernetes for Beginners — browser-based labs, no setup needed
AWS Free Tier (aws.amazon.com/free)	Cloud Platform	Free	12 months of free AWS usage — 750 hours EC2, 5GB S3, 25GB RDS per month
roadmap.sh/devops	Roadmap	Free	Community-built visual DevOps roadmap — the most comprehensive overview available
Terraform Get Started Tutorials (HashiCorp)	Documentation	Free	Official Terraform tutorials — very well written, start with AWS track
GitHub Actions Documentation	Documentation	Free	Official docs — the 'Quickstart' gets you a working pipeline in 10 minutes
The Odin Project — DevOps Section	Course	Free	Solid foundation for Git, Linux, and deployment basics
AWS Skill Builder (Free Tier)	Online Courses	Free	Official AWS learning — free courses for every AWS service
DevOps with Docker (University of Helsinki)	Online Course	Free	Excellent free university course covering Docker and containerization deeply

Paid Resources (Worth the Money)

Resource Name	Type	Cost	Why Use It
KodeKloud Pro Subscription	Platform	~INR 2000/month	The best hands-on DevOps learning platform in the world — labs for every tool
Udemy — DevOps Bootcamp by Mumshad Mannambeth	Course	~INR 500 (sale)	One of the highest-rated DevOps courses — covers Docker, K8s, Ansible, Terraform
A Cloud Guru (Pluralsight)	Platform	~INR 2500/month	Strong AWS certification preparation content with hands-on labs
AWS Solutions Architect Exam (SAA-C03)	Exam	~USD 150 (~INR 12,500)	The most recognized cloud certification in India — doubles starting salary for many

Projects to Build

Project 1: Multi-VM App Deployment

- What you build: Use Vagrant to spin up 3 virtual machines — web server (nginx), app server (Node.js), database (MySQL). Configure all three manually first, then automate with Bash scripts.
- Skills demonstrated: Linux, Vagrant, networking, Bash, service configuration
- Estimated build time: Weekend project (~8 hours)
- Best done in: Phase 1

Project 2: Containerized Microservices App

- What you build: Take a simple web application with a frontend and a backend. Write separate Dockerfiles for each. Use Docker Compose to run them together. Publish images to Docker Hub.
- Skills demonstrated: Docker, Docker Compose, Docker Hub, multi-service architecture
- Estimated build time: 1 week (10-15 hours)
- Best done in: Phase 2

Project 3: Full CI/CD Pipeline

- What you build: Connect a GitHub repository to GitHub Actions. On every push: run automated tests, build a Docker image, push to Docker Hub, deploy to an EC2 instance automatically. Zero manual steps.
- Skills demonstrated: GitHub Actions, Docker, EC2, SSH deployment, environment secrets
- Estimated build time: 1 week (12-18 hours)
- Best done in: Phase 2

Project 4: AWS Infrastructure with Terraform (Capstone)

- What you build: Write Terraform code that creates a complete production-ready environment: VPC with public/private subnets, EC2 auto-scaling group, Application Load Balancer, RDS database, S3 for static files. One command creates everything, one destroys it.
- Skills demonstrated: Terraform, AWS (VPC, EC2, ALB, RDS, S3, IAM), state management
- Estimated build time: 10-14 days (20-30 hours)
- Best done in: Phase 3

Project 5: Kubernetes Monitoring Stack

- What you build: Deploy your containerized app to a local Kubernetes cluster (minikube). Add Prometheus for metrics and Grafana for visualization. Create dashboards showing request rate, error rate, and resource usage. Set up one alert.
- Skills demonstrated: Kubernetes, Helm, Prometheus, Grafana, alerting rules
- Estimated build time: 1 week (10-15 hours)
- Best done in: Phase 3

Career Outcomes

Job Title	Experience	Salary (India)	Primary Skills
Junior DevOps Engineer	0-1 year	INR 6-12 LPA	Linux, Git, CI/CD, Docker basics
Cloud Support Engineer	0-1 year	INR 5-10 LPA	AWS basics, troubleshooting, networking
DevOps Engineer	1-3 years	INR 12-22 LPA	Docker, K8s, Terraform, full CI/CD
Site Reliability Eng.	2-4 years	INR 18-30 LPA	SLOs, incident response, automation
Cloud Engineer	2-4 years	INR 15-28 LPA	Multi-cloud, IaC, cost optimization
Platform Engineer	3-6 years	INR 25-45 LPA	Internal developer platforms, K8s at scale
DevOps Architect	6+ years	INR 40-80 LPA	Enterprise architecture, team leadership

Certifications Worth Getting (in order)

- AWS Cloud Practitioner (CLF-C02) — Free study material, exam ~INR 8,000. Get this first.

- AWS Solutions Architect Associate (SAA-C03) — Most recognized cert, exam ~INR 12,500. Significantly increases salary.
- Certified Kubernetes Administrator (CKA) — Highly respected, exam ~USD 395 (~INR 33,000). For roles above INR 18 LPA.
- HashiCorp Terraform Associate — Validates IaC skills, exam ~USD 70 (~INR 6,000). Good addition.

COURSE 3 OF 5

Frontend Development

One-Line Description

Build the websites and apps that people see and interact with — from your first HTML page to deploying a full React application visible to the whole world.

Course Overview

Field	Detail
Full Name	Frontend Web Development (HTML, CSS, JavaScript, React)
Difficulty	Beginner — No prior experience required. Logical thinking helps.
Prerequisites	Basic computer literacy. No coding experience needed to start.
Duration	90 Days (2-3 hours/day)
Time to Income	60-90 days — freelance projects are possible as early as month 2
Best For	Creative people who like visual work, designers wanting to code, anyone who admires websites
NOT For	People who hate visual design or have no patience for pixel-level details
Salary (Fresher)	INR 3.5-8 LPA (Junior Frontend Developer, UI Developer, Web Developer)
Salary (Mid)	INR 10-20 LPA (Frontend Developer, React Developer)
Salary (Senior)	INR 22-50 LPA (Senior Frontend, Staff Engineer, Frontend Architect)
Top Companies	Flipkart, Meesho, Zomato, Razorpay, Freshworks, startups, agencies

What This Field Actually Is

Frontend development means building everything a user sees and interacts with on a website or web app — the buttons, layouts, forms, animations, navigation menus, and product pages. If you've ever admired how a website looks or felt satisfying to use, a frontend developer built that.

The core technologies: HTML creates the structure (like the skeleton), CSS makes it look good (like the clothes), and JavaScript makes it interactive (like the muscles). React is a JavaScript library that makes building complex, dynamic interfaces much faster and more organized.

In 2026, React is used by approximately 44% of all professional developers worldwide. It is the safest skill to learn for job placement. Companies like Flipkart, Razorpay, Meesho, and virtually every Indian startup use React.

Skills You Will Have After 90 Days

Skill	Level After Course	Real-World Use
HTML5	Intermediate	Structure any webpage with semantic, accessible markup
CSS3 + Flexbox + Grid	Intermediate	Create responsive, beautiful layouts that work on all screen sizes
JavaScript (ES6+)	Intermediate	Add interactivity, handle events, fetch data from APIs, manipulate the DOM
React (Functional Components)	Intermediate	Build complex UIs with components, props, and state
React Hooks	Intermediate	useState, useEffect, useContext, useRef — the core React patterns
React Router	Beginner-Mid	Multi-page apps with client-side routing
Tailwind CSS	Intermediate	Utility-first styling used by most modern startups
REST APIs	Beginner-Mid	Fetch data from backend APIs and display it in the UI
Git & GitHub	Beginner-Mid	Version control and collaboration
Deployment (Vercel/Netlify)	Beginner	Deploy projects live with a public URL in minutes
Responsive Design	Intermediate	Make sites that look great on mobile, tablet, and desktop
TypeScript Basics	Beginner	Type-safe JavaScript — increasingly required in job listings

90-Day Week-by-Week Roadmap

Phase 1: Foundation

HTML, CSS, and JavaScript — the three pillars of the web

Week 1: HTML — Structure of the Web

Daily commitment: 1.5-2 hours/day

- What is HTML and how browsers parse it — the document tree
- Core tags: html, head, body, div, p, h1-h6, a, img, ul/ol/li, span
- Semantic HTML5 tags: header, nav, main, section, article, footer, aside — why these matter for SEO and accessibility
- Forms: input types (text, email, password, checkbox, radio, select), labels, form submission

- HTML attributes: id, class, href, src, alt, placeholder, required
- Project: Build a complete static personal bio page — your name, photo, about section, skills list, and contact form
- Resource: MDN Web Docs HTML Introduction — the definitive free reference

Milestone: You can build any static webpage layout using semantic HTML — your bio page is live on your computer

Week 2: CSS — Making Things Look Good

Daily commitment: 2 hours/day

- Box model: content, padding, border, margin — understanding this is fundamental
- Selectors: element, class, id, descendant, pseudo-classes (:hover, :focus, :nth-child)
- Colors, fonts, backgrounds, borders, shadows — visual styling
- Flexbox: the most important layout tool. Practice with Flexbox Froggy (flexboxfroggy.com) until you beat all 24 levels
- CSS Grid: two-dimensional layouts. Practice with Grid Garden (cssgridgarden.com)
- Responsive design: media queries — @media (max-width: 768px) { ... } — make layouts that adapt to screen size
- CSS variables and custom properties
- Project: Style your HTML bio page to look professional — clean typography, spacing, colors, mobile-responsive
- Resource: Kevin Powell's CSS YouTube channel — the best CSS teacher on the internet, all free

Milestone: You can recreate any website layout using Flexbox and Grid, and your bio page looks professional on both desktop and mobile

Week 3: JavaScript Fundamentals

Daily commitment: 2.5 hours/day

- Variables: const, let — when to use each (always prefer const first)
- Data types: strings, numbers, booleans, null, undefined, arrays, objects
- Functions: regular, arrow functions, default parameters
- Array methods: map, filter, reduce, find, forEach, some, every — these are used constantly in React
- Objects: dot notation, bracket notation, destructuring, spread operator
- DOM manipulation: document.getElementById, querySelector, addEventListener, innerHTML, classList
- Events: click, submit, input, keydown — make web pages respond to user actions
- Project: Build an interactive quiz — 5 questions, user selects answers, score shown at end. Pure HTML/CSS/JS, no libraries.
- Resource: javascript.info — the best free JavaScript tutorial in existence

Milestone: You understand core JavaScript and can add interactivity to any webpage — your quiz app works correctly

Week 4: JavaScript Intermediate & APIs

Daily commitment: 2.5-3 hours/day

- Asynchronous JS: setTimeout, setInterval, Promises, async/await — essential for fetching data
- Fetch API: make HTTP requests to external APIs and display the data
- JSON: parse responses with JSON.parse(), create JSON with JSON.stringify()
- Error handling: try/catch/finally — handle network failures gracefully
- Local storage: save data in the browser that persists after page refresh
- ES6 modules: import and export — how to organize code across multiple files
- Project 1: Weather app — user enters a city, fetches real weather data from OpenWeatherMap API (free key), displays temperature and conditions with icons
- Project 2: GitHub profile viewer — enter a GitHub username, display their repos, bio, and follower count using the GitHub API
- Resource: The Odin Project — JavaScript section. Free, project-based, excellent quality.

Milestone: You can build web apps that fetch real data from external APIs and display it dynamically

Phase 2: React & Modern Tools

Build real-world apps with React, the industry-standard frontend library

Week 5: React Fundamentals

Daily commitment: 2.5 hours/day

- Why React: the problem with plain JS as apps get complex — and how React solves it
- Create a React project: npx create-react-app my-app OR npx create vite@latest my-app (use Vite)
- JSX: HTML inside JavaScript — the syntax React uses
- Components: the building block of React apps — functional components only
- Props: pass data from parent to child components
- useState hook: how React manages changing data — build a counter, a toggle, a form
- Conditional rendering: show/hide UI based on state — {isLoggedIn && <Dashboard />}
- Rendering lists: .map() over arrays, the key prop importance
- Project: Build a Todo app in React — add, delete, and mark tasks as complete. Classic beginner project, teaches all the fundamentals.

Milestone: You can build a functional React application from scratch using components, props, and state

Week 6: React — Hooks & API Integration

Daily commitment: 3 hours/day

- useEffect hook: run side effects (API calls, subscriptions) when component mounts or state changes
- Fetch data in React: useEffect + fetch + useState to load and display API data
- Loading and error states: show a spinner while data loads, show an error message if it fails
- Forms in React: controlled components — form input synced with state via onChange
- useRef hook: access DOM elements directly, persist values without re-rendering
- Custom hooks: extract repeated logic into reusable functions (e.g., useFetch, useLocalStorage)

- Tailwind CSS with React: install tailwindcss and use utility classes — no more custom CSS files
- Project: Movie search app — search the OMDb API (free), display results as cards with poster, title, year, and rating. Add a 'favorites' feature using localStorage.

Milestone: You can build a complete data-driven React app with API integration, loading states, and form handling

Week 7: React Router & State Management

Daily commitment: 3 hours/day

- React Router v6: multi-page apps without page reloads — <BrowserRouter>, <Routes>, <Route>, <Link>
- Dynamic routes: /user/:id — pass data through URL parameters
- Protected routes: redirect unauthenticated users to login page
- useNavigate, useParams, useLocation hooks — programmatic navigation
- Context API: share state across many components without prop drilling
- Introduction to Zustand (lightweight state management): cleaner alternative to Redux for beginners
- Project: Build a multi-page e-commerce product catalog — homepage with product grid, product detail page, shopping cart with item count in navbar, all using React Router and Context API for cart state

Milestone: You can build a multi-page React application with client-side routing and global state management

Week 8: Portfolio & Deployment

Daily commitment: 2.5 hours/day

- Build your portfolio website in React — professional, responsive, shows all 4 projects
- Deploy on Vercel: connect GitHub repo, automatic deployment on every push — free forever for personal projects
- Performance basics: why slow websites lose users, lazy loading images, code splitting in React (React.lazy)
- TypeScript introduction: add types to your React components — any[], string, interface UserProps
- Introduction to Next.js: React with server-side rendering — the production-grade framework
- Git workflow for teams: feature branches, pull requests, code review process
- Accessibility basics: alt text, ARIA labels, semantic HTML — your apps should work with screen readers
- SEO basics: meta tags, Open Graph tags, page titles — help your portfolio rank on Google

Milestone: Your portfolio website is live on Vercel with a custom URL, all 4 projects are deployed and accessible

Phase 3: Job Ready

Polish portfolio, apply strategically, ace the technical interview

Week 9: Advanced Patterns & Interview Prep

Daily commitment: 2-2.5 hours/day

- React performance: useMemo, useCallback — when and why to use them
- Common interview questions: what is the virtual DOM, explain React reconciliation, what are hooks, difference between useMemo and useCallback
- Practice 5 frontend LeetCode problems: Two Sum, Valid Parentheses, Reverse String, FizzBuzz, Palindrome Check — these come up in JavaScript rounds
- CSS interview questions: flexbox vs grid, position absolute vs relative, how specificity works, what is the box model
- Build a clone: reproduce a famous website's UI (Netflix landing page, Airbnb search page, Swiggy homepage) — this shows you can work to a design

Milestone: You can answer the 20 most common frontend interview questions confidently

Week 10: Job Applications

Daily commitment: 3+ hours/day

- Apply to 5-10 roles per day: search 'Frontend Developer React', 'React Developer', 'UI Developer' on LinkedIn, Naukri, Instahyre
- Freshers target: companies with 50-500 employees — they are more likely to hire and train juniors
- For freelancing: Upwork profile titled 'React Developer | Responsive Web Apps | Tailwind CSS'
- Cold outreach on LinkedIn: connect with engineering managers and frontend leads at target companies — send a genuine note with your portfolio link
- Local businesses (Stackwise model): approach shops, restaurants, clinics who need websites — offer your services at INR 10,000-25,000 for a simple responsive site
- Follow up on every application after 5 business days — most people do not follow up

Milestone: 50+ applications sent, portfolio shared with recruiters, first interview calls coming in

Week 11: Mock Interviews & Technical Test Prep

Daily commitment: 3+ hours/day

- Frontend take-home assignments: companies often give you 2-3 days to build a small feature or fix a bug — practice this format
- Live coding prep: practice on CodeSandbox (browser-based React environment) to simulate live coding without local setup
- UI challenges: recreate UI components from a Figma design — many companies test this
- Mock system design: 'How would you build the frontend for Instagram?' — component tree, state management, API integration, pagination
- Review your projects thoroughly — interviewers will ask you to walk through code you wrote
- Research target companies: what tech stack do they use, what are their products, read their engineering blog

Milestone: You have done 5+ mock interviews and can confidently code a feature live in an interview setting

Week 12: First Job or Client

Daily commitment: 1-2 hours/day

- Negotiate salary: entry-level React developers in India earn INR 4-8 LPA. Know the range before accepting.
- For freelancing: finish your first project, collect a testimonial, ask for a referral
- Set up invoicing: use Zoho Invoice (free) or a simple invoice template for freelance payments
- Learn the workplace: how to read existing code, how to ask good questions, how to use git in a team
- 30-60-90 day plan for your first job: week 1 onboard, month 1 contribute small fixes, month 2 own a feature, month 3 lead a small task
- Keep learning: JavaScript is constantly evolving — follow javascript.info, CSS-Tricks, and Smashing Magazine

Milestone: You have accepted a job offer or completed your first paid freelance project

Free Resources

Resource Name	Type	Cost	Why Use It
The Odin Project (theodinproject.com)	Full Curriculum	Free	The most comprehensive free full-stack curriculum — covers HTML/CSS/JS/React from scratch with real projects
freeCodeCamp Responsive Web Design	Online Course	Free	300-hour certificate course — well-structured, covers HTML and CSS thoroughly with projects
javascript.info	Website	Free	The definitive JavaScript reference — clear, comprehensive, regularly updated
Kevin Powell YouTube	YouTube Channel	Free	Best CSS teacher on the internet — Flexbox, Grid, animations, responsive design all explained brilliantly
Official React Documentation (react.dev)	Documentation	Free	Completely redesigned in 2023 — the interactive tutorials are excellent for beginners
Scrimba Frontend Career Path	Interactive Course	Free (limited)	Browser-based coding — edit code inside the video. Very effective for beginners.
Josh W Comeau CSS blog (joshwcomeau.com)	Blog	Free	The most beautifully explained CSS tutorials you will find anywhere — interactive and visual
Flexbox Froggy (flexboxfroggy.com)	Game	Free	Learn Flexbox by helping a frog — finish all 24 levels before moving on
CSS Grid Garden (cssgridgarden.com)	Game	Free	Learn CSS Grid by growing carrots — complete all levels for grid mastery

Resource Name	Type	Cost	Why Use It
Traversy Media YouTube	YouTube Channel	Free	Brad Traversy's channel — crash courses for HTML, CSS, JS, React that are well-organized and practical

Paid Resources (Worth the Money)

Resource Name	Type	Cost	Why Use It
Scrimba Frontend Developer Career Path (Pro)	Course	~INR 1500/month	The interactive coding format is uniquely effective — you code inside the tutorial video
Zero to Mastery — Complete Web Developer	Course	~INR 500 Udemy	Comprehensive bootcamp-style course, well-paced for beginners, covers React thoroughly
Frontend Masters — React Learning Path	Platform	~USD 39/month	Professional-grade courses used by engineers at top companies — if you can afford it

Projects to Build

Project 1: Responsive Portfolio Website

- What you build: A personal portfolio with sections: Hero with your name/tagline, About Me, Skills grid, Projects showcase (4 cards with links), and Contact form. Must be fully responsive on mobile.
- Skills demonstrated: HTML5, CSS3, Flexbox/Grid, media queries, CSS animations
- Estimated build time: 3-5 days
- Best done in: Phase 1

Project 2: Weather Dashboard App

- What you build: User types a city name, app shows current weather (temperature, conditions, humidity, wind speed), a 5-day forecast, and automatically detects user's location on load.
- Skills demonstrated: JavaScript, Fetch API, OpenWeatherMap API, async/await, CSS animations
- Estimated build time: 3-5 days
- Best done in: Phase 1-2

Project 3: Movie Search & Favorites App (React)

- What you build: Search any movie via the OMDb API (free key). Display results as cards. Users can click a heart to save favorites. Favorites persist in localStorage. Filter by genre. Dark mode toggle.

- Skills demonstrated: React, useState, useEffect, API integration, localStorage, Tailwind CSS
- Estimated build time: 5-7 days
- Best done in: Phase 2

Project 4: E-Commerce Product Catalog (React)

- What you build: Homepage with product grid, filter by category and price range, product detail page, add to cart functionality, cart sidebar with item count. Uses React Router for navigation and Context API for cart state.
- Skills demonstrated: React, React Router, Context API, Tailwind CSS, REST API, responsive design
- Estimated build time: 1-2 weeks
- Best done in: Phase 2-3

Project 5: Full Portfolio with Next.js (Capstone)

- What you build: Rebuild your portfolio as a Next.js app. Add a blog section that pulls posts from a markdown folder or a headless CMS. Implement SEO with metadata API. Deploy on Vercel with a custom domain.
- Skills demonstrated: Next.js, TypeScript basics, MDX/markdown, SEO, Vercel deployment
- Estimated build time: 1-2 weeks
- Best done in: Phase 3

Career Outcomes

Job Title	Experience	Salary (India)	Skills Focus
Junior Frontend Developer	0-1 year	INR 3.5-8 LPA	HTML, CSS, JS, basic React
UI Developer	0-1 year	INR 4-7 LPA	Design implementation, pixel-perfect CSS
React Developer	1-3 years	INR 10-18 LPA	React, state management, API integration
Frontend Engineer	2-4 years	INR 15-25 LPA	Performance, testing, architecture
Senior Frontend Engineer	4-7 years	INR 22-40 LPA	System design, mentoring, tech decisions
Frontend Architect	7+ years	INR 40-70 LPA	Architecture, design systems, org-wide impact
Freelance Developer	6+ months	INR 500-3000/hr	Client projects, websites, web apps

COURSE 4 OF 5

Data Analytics

One-Line Description

Turn raw data into actionable business insights — master Excel, SQL, Python, and Power BI to analyze real-world datasets and drive decisions at any company.

Course Overview

Field	Detail
Full Name	Data Analytics (Excel, SQL, Python, Power BI)
Difficulty	Beginner — Commerce, science, or engineering graduates find this very accessible
Prerequisites	Comfort with Excel, basic math (averages, percentages), logical thinking
Duration	90 Days (2-3 hours/day)
Time to Income	90 days — entry-level analyst roles, internships, and freelance data projects
Best For	Commerce grads, MBA aspirants, business-curious people, anyone who loves patterns in data
NOT For	People who hate math or numbers entirely — data requires numerical comfort
Salary (Fresher)	INR 3.5-7 LPA (Junior Data Analyst, Business Analyst, Operations Analyst)
Salary (Mid)	INR 8-15 LPA (Data Analyst, BI Developer, Analytics Engineer)
Salary (Senior)	INR 15-30+ LPA (Senior Analyst, Analytics Manager, Data Science transition)
Top Companies	Ola, Flipkart, Razorpay, consulting firms, BFSI companies, any data-using business

What This Field Actually Is

A data analyst is someone who collects raw data, cleans it, analyzes it for patterns, and presents findings to help a business make better decisions. They answer questions like: 'Why did our revenue drop last month?', 'Which customer segments are most profitable?', 'Which city has the highest order cancellation rate?'

Every company — from a startup to an FMCG giant — generates massive amounts of data daily and desperately needs people who can make sense of it. Data analytics is one of the most

domain-versatile careers: you can work in e-commerce, banking, healthcare, logistics, edtech, or any industry that uses data.

The tool stack: Excel for quick analysis, SQL for querying databases (where most business data lives), Python for large-scale analysis and automation, and Power BI or Tableau for building visual dashboards that non-technical stakeholders can understand.

Skills You Will Have After 90 Days

Skill	Level After Course	Real-World Use
Microsoft Excel / Google Sheets	Advanced	VLOOKUP, Pivot Tables, Power Query, data cleaning, dashboards — used daily by analysts everywhere
SQL	Intermediate	Query any relational database: SELECT, JOIN, GROUP BY, subqueries, CTEs, window functions
Python (Pandas & NumPy)	Beginner-Mid	Clean messy datasets, merge tables, calculate statistics, automate repetitive analysis tasks
Data Visualization	Intermediate	Matplotlib, Seaborn in Python — create charts that tell a clear story
Power BI	Intermediate	Build interactive dashboards connecting to live data sources — the #1 BI tool in India
Statistics Basics	Beginner-Mid	Mean, median, standard deviation, correlation, hypothesis testing, A/B testing basics
Data Cleaning	Intermediate	Handle missing values, outliers, duplicates, inconsistent formats — 80% of real analyst work
Exploratory Data Analysis (EDA)	Intermediate	Systematically explore a new dataset: distributions, correlations, anomalies
Business Communication	Intermediate	Translate analysis into clear recommendations for non-technical stakeholders
Kaggle Competitions	Beginner	Participate in data challenges — builds portfolio credibility

90-Day Week-by-Week Roadmap

Phase 1: Foundation

Excel mastery and SQL — the core tools of every working analyst

Week 1: Excel Mastery — The Analyst's First Tool

Daily commitment: 2 hours/day

- Data types, sorting, filtering, basic formatting in Excel
- Essential formulas: SUM, AVERAGE, COUNT, COUNTA, IF, COUNTIF, SUMIF
- VLOOKUP and XLOOKUP: look up values across tables — used constantly in real analyst work
- INDEX + MATCH: the more flexible alternative to VLOOKUP
- Pivot Tables: summarize and analyze data with zero code — the most powerful Excel feature
- Pivot Charts: visualize pivot table data instantly
- Data validation: prevent bad data entry with dropdown lists and constraints
- Practice dataset: Download a real Indian e-commerce dataset from Kaggle, analyze it entirely in Excel
- Resource: ExcelJet (exceljet.net) — the best free Excel formula reference

Milestone: You can analyze a 10,000-row dataset in Excel using Pivot Tables, VLOOKUP, and complex formulas

Week 2: Advanced Excel & Power Query

Daily commitment: 2 hours/day

- Conditional formatting: color-code cells based on values — instant visual data exploration
- Data cleaning in Excel: remove duplicates, trim whitespace, fix inconsistent text cases
- Power Query: connect to external data sources (CSV, databases, web), transform data without formulas
- Power Query steps: remove columns, filter rows, split columns, merge queries
- Named ranges and dynamic ranges: make formulas easier to read and maintain
- Excel dashboards: combine charts, slicers, and conditional formatting into a management dashboard
- Project: Build a sales performance dashboard in Excel using a real dataset — monthly trends, top products, regional breakdown, with slicers for filtering

Milestone: You can build a professional Excel dashboard with Power Query, dynamic charts, and slicers that a business can actually use

Week 3: SQL Fundamentals

Daily commitment: 2 hours/day

- Install MySQL (free) or use SQLiteOnline.com (browser-based, zero setup)
- SELECT, FROM, WHERE — the foundation of every SQL query
- ORDER BY, LIMIT — sort and paginate results
- Aggregate functions: COUNT(), SUM(), AVG(), MIN(), MAX()
- GROUP BY: summarize data by category — 'total revenue by city', 'average order value by month'
- HAVING: filter after GROUP BY — 'only cities with revenue above 1 crore'
- String functions: UPPER, LOWER, TRIM, CONCAT, SUBSTRING — clean text data in SQL

- Date functions: YEAR(), MONTH(), DATEDIFF() — analyze time-based data
- Practice: Alex the Analyst's SQL Tutorial on YouTube — free, practical, beginner-friendly
- Practice dataset: Use the Northwind database — a classic dataset with orders, products, customers

Milestone: You can write SQL queries to answer business questions from any relational database

Week 4: SQL Intermediate & Advanced

Daily commitment: 2.5 hours/day

- JOINS: INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN — combine data from multiple tables
- Understand the difference: LEFT JOIN keeps all rows from the left table even if no match
- Subqueries: write a query inside another query — find customers who ordered more than the average
- CTEs (Common Table Expressions): WITH clause — make complex queries readable
- Window functions: ROW_NUMBER(), RANK(), DENSE_RANK(), LAG(), LEAD() — for advanced analysis
- CASE WHEN: conditional logic inside SQL — create segments, categories, labels
- Data cleaning with SQL: handling NULLs (IS NULL, COALESCE), deduplication
- Project: Complete a SQL data exploration project — analyze a real dataset (Netflix shows, Spotify data, Zomato orders), answer 10 business questions, document findings
- Resource: Mode Analytics SQL Tutorial (free) — interactive browser-based practice

Milestone: You can write complex multi-table SQL queries using JOINs, CTEs, and window functions

Phase 2: Python & Visualization

Analyze data at scale with Python, Pandas, and Power BI

Week 5: Python for Data Analysis — Basics

Daily commitment: 2.5 hours/day

- Install Python 3.11 and Jupyter Notebook (easiest: install Anaconda, which includes everything)
- Python basics specific to data analysis: variables, lists, dictionaries, loops, functions
- NumPy: arrays, array operations, broadcasting, mathematical functions
- Pandas introduction: DataFrames — think of it as Excel but in Python, with superpowers
- Reading data: pd.read_csv(), pd.read_excel() — load datasets in one line
- Exploring data: df.head(), df.info(), df.describe(), df.shape, df.columns
- Selecting and filtering: df['column'], df[df['price'] > 100], df.loc[], df.iloc[]
- Practice: Load a Kaggle dataset into Pandas, explore its basic statistics and structure
- Resource: Kaggle's free Pandas course — 4 hours, excellent practical exercises

Milestone: You can load, inspect, and do basic filtering on any dataset using Pandas in a Jupyter Notebook

Week 6: Python — Data Cleaning & EDA

Daily commitment: 2.5 hours/day

- Missing data: df.isnull().sum(), df.dropna(), df.fillna() — handle missing values strategically
- Duplicate data: df.duplicated(), df.drop_duplicates()
- Data type conversion: df['price'].astype(float), pd.to_datetime()
- String operations in Pandas: .str.upper(), .str.strip(), .str.replace(), .str.contains()
- Groupby in Pandas: df.groupby('city')['revenue'].sum() — same concept as SQL GROUP BY
- Merge and Join: pd.merge(df1, df2, on='customer_id') — same concept as SQL JOINs
- Exploratory Data Analysis: systematic process — shape, distributions, correlations, outliers
- Project: Take a messy real-world dataset (Indian startup funding, IPL player stats, Zomato restaurant data), clean it completely, and produce a 10-insight EDA report in a Jupyter Notebook

Milestone: You can take a completely messy, real-world dataset and produce a clean, analyzed Jupyter Notebook with 10 business insights

Week 7: Data Visualization — Charts That Tell Stories

Daily commitment: 2.5 hours/day

- Matplotlib: the foundation of Python visualization — line, bar, scatter, histogram, pie charts
- Seaborn: beautiful statistical visualizations — heatmaps, pair plots, box plots, violin plots built on Matplotlib
- Chart selection guide: when to use each chart type (bar for comparison, line for trends, scatter for correlation, heatmap for correlation matrices)
- Storytelling with data: title, axis labels, annotations, color choice — make the insight obvious
- Power BI introduction: install Power BI Desktop (free on Windows), connect to a CSV or Excel file
- Power BI basics: load data, create relationships between tables, build bar charts, line charts, and card visuals
- Power BI slicers and filters: make dashboards interactive without code
- Project: Recreate your Python EDA visualizations in Power BI — build an interactive dashboard with 5+ visuals, slicers for country/year/category, and a KPI card section

Milestone: You can choose the right chart type, create clear visualizations in Python, and build an interactive Power BI dashboard

Week 8: Statistics & Business Analytics

Daily commitment: 3 hours/day

- Descriptive statistics: mean, median, mode, standard deviation, variance, percentiles — interpret each one
- Distributions: normal distribution, right-skewed, left-skewed — recognize them in your data
- Correlation: Pearson correlation coefficient, correlation matrix heatmap — 'does price affect sales?'

- Introduction to A/B testing: hypothesis testing, p-values, statistical significance — used in product and marketing analytics
- Cohort analysis: track user groups over time — classic e-commerce and SaaS analysis
- Funnel analysis: where do users drop off in your purchase or signup flow?
- Business metrics: DAU/MAU, CAC, LTV, churn rate, NPS — every analyst must know these cold
- Project: Complete a business analytics case study — take a company's dataset (Instacart, Olist e-commerce, or any Kaggle business dataset), answer specific business questions using statistics, create a PowerPoint presentation with findings

Milestone: You can perform statistical analysis and present business insights in a structured, professional format

Phase 3: Portfolio & Career

Build a strong data portfolio and land your first analyst role

Week 9: Portfolio Building

Daily commitment: 3 hours/day

- Choose 3 project domains to showcase: pick industries you're interested in (fintech, e-commerce, healthcare, sports)
- Build Project 1: SQL + Python + Power BI end-to-end project on a business dataset
- Host projects on Kaggle (easiest — submit notebooks, get community feedback)
- Create a GitHub account: upload Python notebooks with clear documentation
- Portfolio website on Notion or GitHub Pages: list your projects with screenshots, tools used, and key findings
- Write 2 LinkedIn posts about your analysis projects — 'I analyzed 10,000 Zomato orders, here's what I found'

Milestone: You have 3 data analysis projects published publicly on Kaggle and GitHub

Week 10: Job Applications

Daily commitment: 3+ hours/day

- Apply to 5-10 roles/day: search 'Data Analyst', 'Business Analyst', 'Operations Analyst', 'BI Analyst' on LinkedIn and Naukri
- Entry-level friendly companies: consulting firms (Deloitte, KPMG, PwC), e-commerce companies, BFSI (banks, insurance, fintech)
- Tailor resume for each role: list specific tools mentioned in the job description
- Prepare for Excel tests: many analyst roles include a 30-60 minute Excel or SQL assessment at the start
- Google Data Analytics Certificate (Coursera, 6 months part-time): recognized by many Indian employers, costs ~INR 3500/month
- Reach out to analysts at target companies: ask for a 15-minute call about their day-to-day experience

Milestone: 50+ applications sent, at least 10 companies progressed to assessment stage

Week 11: Interview & Assessment Prep

Daily commitment: 3+ hours/day

- SQL interview prep: practice on HackerRank SQL (free) — medium difficulty problems test JOIN and GROUP BY skills
- Excel assessments: be ready to build a Pivot Table, write a VLOOKUP, and clean data in a timed test
- Case study interviews: 'Our checkout conversion dropped 15% last month. Walk me through how you'd investigate this.' — practice this format
- Behavioral questions for analysts: 'Tell me about a time you found an insight that changed a decision' — prepare 3 STAR-method answers
- Statistics questions: explain the difference between mean and median, when to use each, what an outlier is
- Power BI/Tableau: be ready to share your dashboard and walk through design decisions

Milestone: You can pass a standard SQL assessment, present a data case study confidently, and answer common analyst interview questions

Week 12: Specialize & Continue Learning

Daily commitment: 1-2 hours/day

- Pick a domain specialization: fintech analytics, product analytics, marketing analytics, or supply chain analytics
- Learn Power BI DAX: calculated columns, measures, time intelligence functions — this makes you more valuable
- Introduction to basic machine learning: linear regression — when does a data analyst become a data scientist?
- Microsoft PL-300 (Power BI Data Analyst Associate) certification: very recognized in India, exam ~INR 12,000
- Google Data Analytics Professional Certificate (Coursera): widely recognized, 6 months part-time, ~INR 3500/month
- Join data communities: r/dataanalysis, r/BusinessIntelligence, LinkedIn data analytics groups
- First 90 days in the job: over-communicate your findings, ask 'so what?' before presenting any analysis

Milestone: You are settled in your first analyst role and have a clear 12-month growth plan

Free Resources

Resource Name	Type	Cost	Why Use It
Alex the Analyst YouTube	YouTube Channel	Free	The best YouTube channel for data analysts — SQL, Python, Excel, Power BI all covered practically
Kaggle (kaggle.com)	Platform	Free	Free datasets, free Python/SQL courses, competitions to build portfolio — the data analyst's home base
Mode Analytics SQL Tutorial	Interactive	Free	Browser-based SQL practice with real

Resource Name	Type	Cost	Why Use It
			business questions — excellent for intermediate SQL
SQL Murder Mystery (sql-murder-mystery.com)	Game	Free	Solve a murder mystery using SQL — surprisingly effective for learning JOINs and subqueries
freeCodeCamp Data Analysis with Python	YouTube/ Course	Free	Complete video course covering Pandas, NumPy, Matplotlib, and real data projects
Power BI Documentation (Microsoft)	Documentation	Free	Official Microsoft docs — the Guided Learning section is excellent for beginners
StatQuest with Josh Starmer (YouTube)	YouTube Channel	Free	The best statistics explainer on the internet — makes complex statistical concepts simple and visual
roadmap.sh/data-analyst	Roadmap	Free	Visual roadmap of every skill a data analyst needs — good for knowing what to learn next
W3Schools SQL Tutorial	Website	Free	Quick SQL reference and exercises — good for reviewing specific syntax
DataCamp (Free Tracks)	Online Courses	Free (limited)	Interactive browser-based courses in Python and SQL — free tier has enough to get started

Paid Resources (Worth the Money)

Resource Name	Type	Cost	Why Use It
Google Data Analytics Certificate (Coursera)	Certificate	~INR 3500/month	8-course program by Google, very well-structured, recognized by employers, includes discount for CompTIA exam
Udemy — Data Analyst Bootcamp by Alex Freberg	Course	~INR 500 (sale)	Alex the Analyst's own course — consistent with his YouTube content, great for portfolio projects
Microsoft PL-300 Power BI Exam Prep (Udemy)	Course	~INR 500 (sale)	Well-structured Power BI prep — the PL-300 exam is recognized in Indian corporate environments
DataCamp Data Analyst Career Track	Course Track	~INR 2500/month	Comprehensive interactive track — if you prefer learning by doing in a browser over videos

Projects to Build

Project 1: Sales Performance Excel Dashboard

- What you build: Create a comprehensive sales analysis dashboard using a real retail dataset. Include: monthly revenue trend, top 10 products by sales, regional heatmap, YoY comparison, and an executive summary slide.
- Skills demonstrated: Excel Power Query, Pivot Tables, dynamic charts, conditional formatting, slicers
- Estimated build time: 3-5 days
- Best done in: Phase 1

Project 2: SQL Business Investigation

- What you build: Download a business database (Northwind, Chinook, or a Kaggle SQL dataset). Write 15+ SQL queries answering real business questions: top customers, product performance, seasonal trends, cohort retention.
- Skills demonstrated: SQL (JOINs, GROUP BY, CTEs, window functions), data storytelling
- Estimated build time: 3-5 days
- Best done in: Phase 1-2

Project 3: Python EDA Report — Indian Dataset

- What you build: Take an India-specific dataset (IPL stats, Bollywood box office, Zomato restaurant data, startup funding data — all on Kaggle). Perform complete EDA: data cleaning, univariate and bivariate analysis, correlation, and 10+ visualizations.
- Skills demonstrated: Python, Pandas, Seaborn, Matplotlib, Jupyter Notebook, statistical analysis
- Estimated build time: 5-7 days
- Best done in: Phase 2

Project 4: Power BI Business Intelligence Dashboard

- What you build: Connect Power BI to a multi-table dataset. Create a dashboard with: 6+ visual types, calculated measures using DAX, time intelligence (MTD, YTD, previous year comparison), and a navigation page with drill-through.
- Skills demonstrated: Power BI, DAX, data modeling, dashboard design, storytelling
- Estimated build time: 5-7 days
- Best done in: Phase 2-3

Project 5: End-to-End Analytics Project (Capstone)

- What you build: Pick a real business question in a domain you care about. Collect or download the data, clean with Python, analyze with SQL, build a Power BI dashboard, and write a 5-page report with executive summary, methodology, findings, and recommendations. Present it as if to a real stakeholder.
- Skills demonstrated: All tools: SQL, Python, Power BI, Excel, business communication
- Estimated build time: 2-3 weeks
- Best done in: Phase 3

Career Outcomes

Job Title	Experience	Salary (India)	Key Tools
Junior Data Analyst	0-1 year	INR 3.5-6 LPA	Excel, SQL, basic Python
Business Analyst	0-2 years	INR 5-10 LPA	Excel, SQL, PowerPoint, stakeholder comms
Data Analyst	1-3 years	INR 8-15 LPA	SQL, Python, Power BI, statistics
BI Developer	2-4 years	INR 10-18 LPA	Power BI/Tableau, DAX, data modeling
Product Analyst	2-4 years	INR 12-22 LPA	SQL, Python, product metrics, A/B testing
Analytics Engineer	3-5 years	INR 15-28 LPA	SQL, dbt, Python, data pipelines
Data Analytics Manager	5+ years	INR 22-40 LPA	Team management, strategy, stakeholder mgmt

COURSE 5 OF 5

Cybersecurity & Ethical Hacking

One-Line Description

Learn to defend systems and legally hack into them to find vulnerabilities before malicious attackers do — from networking fundamentals to penetration testing and bug bounty hunting.

Course Overview

Field	Detail
Full Name	Cybersecurity & Ethical Hacking
Difficulty	Intermediate — requires patience and a systematic, methodical mindset
Prerequisites	Basic networking understanding (what an IP address is), comfort with computers, some Linux exposure helpful
Duration	90 Days (2-3 hours/day) to reach entry-level readiness
Time to Income	90 days to SOC Analyst or junior security roles; bug bounties possible in Month 3
Best For	Detail-oriented problem solvers, people who think like attackers, CS grads curious about security
NOT For	People who want quick results — cybersecurity requires deep understanding and patience
Salary (Fresher)	INR 4-8 LPA (SOC Analyst, Junior Pentester, IT Security Analyst)
Salary (Mid)	INR 10-20 LPA (Security Analyst, Penetration Tester, Security Engineer)
Salary (Senior)	INR 25-60+ LPA (CISO, Security Architect, Principal Security Engineer)
Top Companies	Wipro, HCL, Infosys security practices, banks, MSSPs, global security firms

What This Field Actually Is

Cybersecurity professionals protect digital systems, networks, and data from unauthorized access, damage, and theft. There are two sides: the defenders (Blue Team) who monitor, detect, and respond to attacks, and the attackers (Red Team / ethical hackers / penetration testers) who are paid to attack systems legally to find vulnerabilities before criminals do.

The global cybersecurity workforce shortage exceeds 4 million professionals. In India specifically, every bank, hospital, government agency, and tech company is actively hiring security talent. The field offers exceptional job security because the threat never stops growing.

This course takes you through the defender's foundation (networking, operating systems, security concepts) and then into offensive techniques (ethical hacking, penetration testing, CTF challenges). By day 90, you can perform basic web application penetration tests and work as a SOC analyst.

Skills You Will Have After 90 Days

Skill	Level After Course	Real-World Use
Networking Fundamentals	Intermediate	Understand how attacks travel through networks, set up network security controls
Linux for Security	Intermediate	Navigate Kali Linux, use security tools from the command line
Windows Security Basics	Beginner	Understand Active Directory, Windows event logs, Group Policy
OWASP Top 10	Intermediate	Identify and test for the 10 most critical web vulnerabilities
Network Scanning (Nmap)	Intermediate	Discover hosts, open ports, and services on a network
Burp Suite	Beginner-Mid	Intercept and modify HTTP traffic to find web vulnerabilities
Metasploit Framework	Beginner	Exploit known vulnerabilities in controlled lab environments
Wireshark	Beginner-Mid	Capture and analyze network traffic to detect anomalies
TryHackMe / Hack The Box	Intermediate	Complete CTF (Capture The Flag) challenges — your practical portfolio
Python for Security	Beginner	Write scripts that automate security tasks — port scanners, brute force tools
Incident Response Basics	Beginner	Respond to a security incident: containment, investigation, remediation
Report Writing	Beginner-Mid	Document vulnerabilities found during a pentest in a professional format

90-Day Week-by-Week Roadmap

Phase 1: Foundation

Networking, Linux, and security fundamentals — the ground floor of all cybersecurity

Week 1: Networking Fundamentals

Daily commitment: 2 hours/day

- OSI model: 7 layers and what happens at each — physical, data link, network, transport, session, presentation, application
- TCP/IP model: the practical 4-layer model used in real networks
- IP addressing: IPv4 vs IPv6, public vs private IPs, subnetting basics
- DNS: how domain names resolve to IP addresses, DNS records (A, MX, CNAME, TXT)
- HTTP/HTTPS: request/response cycle, status codes (200, 301, 404, 500), headers, cookies, sessions
- Common ports to memorize: 21 (FTP), 22 (SSH), 25 (SMTP), 53 (DNS), 80 (HTTP), 443 (HTTPS), 3389 (RDP), 3306 (MySQL)
- Tools: ping, traceroute, nslookup, netstat — run these and understand the output
- Resource: Professor Messer's CompTIA Network+ free video series on YouTube — the clearest networking explainer

Milestone: You can explain the OSI model, TCP/IP model, and how a web request travels from browser to server and back

Week 2: Linux for Security

Daily commitment: 2 hours/day

- Install Kali Linux on VirtualBox (free) — Kali is the security professional's operating system, comes with 600+ security tools
- Linux command line review: file navigation, permissions, processes, networking commands
- File system structure: /etc (configs), /var/log (logs), /home (users) — know what lives where
- User management: useradd, passwd, sudo — understand privilege escalation concepts
- Network commands: ifconfig/ip, netstat, ss, route, curl, wget
- Service management: systemctl start/stop/status apache2 — manage and inspect running services
- Log analysis: /var/log/auth.log, /var/log/syslog — read logs to detect suspicious activity
- Bash scripting basics for security: automate repetitive security tasks (port scan, file integrity check)
- Resource: OverTheWire Bandit (overthewire.org/wargames/bandit) — learn Linux by solving hacking puzzles, completely free

Milestone: You are comfortable in a Linux terminal and can perform basic system administration tasks relevant to security

Week 3: Security Fundamentals & Core Concepts

Daily commitment: 2 hours/day

- The CIA Triad: Confidentiality, Integrity, Availability — the foundation of all security thinking
- Authentication vs Authorization: identifying who you are vs what you can do

- Common attack types: phishing, malware (viruses, ransomware, trojans, spyware), MITM, DDoS, SQL injection, XSS
- Cryptography basics: symmetric (AES) vs asymmetric (RSA) encryption, hashing (SHA-256, MD5), digital signatures
- PKI: Public Key Infrastructure — how HTTPS certificates work
- Firewalls: stateful vs stateless, iptables basics on Linux
- VPNs: how they work and why they are not as secure as most people think
- SIEM concept: Security Information and Event Management — what SOC analysts use to monitor threats
- Start CompTIA Security+ study: it is the most recognized entry-level certification. Free study material on Professor Messer's site.

Milestone: You understand core security concepts and can explain common attack types and defenses to a non-technical person

Week 4: Ethical Hacking Methodology

Daily commitment: 2.5 hours/day

- The ethical hacking phases: Reconnaissance → Scanning → Gaining Access → Maintaining Access → Covering Tracks
- IMPORTANT LEGAL NOTE: Always get written permission before testing any real system. Practice only on your own machines or designated platforms (TryHackMe, Hack The Box).
- Reconnaissance: passive (Google dorking, WHOIS, Shodan, LinkedIn) vs active (Nmap, direct probing)
- Google Dorking: advanced Google searches that reveal sensitive files, login pages, and vulnerable systems
- Shodan (shodan.io): the search engine for internet-connected devices — legally explore exposed systems
- Sign up for TryHackMe (free tier available): complete the 'Pre-Security' path — networking, Linux, web fundamentals
- Complete your first 5 TryHackMe rooms: Intro to Linux, Intro to Networking, Intro to Web Hacking, Intro to Cybersecurity, DNS in Detail
- Buy TryHackMe Premium if possible (~INR 750/month) — gives full access to learning paths and all rooms

Milestone: You understand the ethical hacking methodology and have completed 5 TryHackMe rooms — your first hands-on security experience

Phase 2: Offensive Security

The tools and techniques of ethical hacking — web app testing, network attacks, and exploitation

Week 5: Network Scanning & Enumeration

Daily commitment: 2.5 hours/day

- Nmap: the most important security tool — learn every flag and what it does
- nmap -sV: service version detection — what software is running on each port?
- nmap -sC: run default scripts — quickly identify common misconfigurations

- nmap -O: OS detection — what operating system is the target running?
- Aggressive scan: nmap -A -T4 (use only in your own lab — this is noisy and detectable)
- Netdiscover: discover live hosts on a local network
- Gobuster/Dirbuster: brute force hidden directories and files on web servers
- Nikto: basic web server vulnerability scanner — find common misconfigurations
- Practice: Set up a vulnerable VM (Metasploitable2 — free download) in VirtualBox and scan it with Nmap. Enumerate every open port.

Milestone: You can perform a complete network scan on a target machine, enumerate all services, and produce a structured report of your findings

Week 6: Web Application Hacking — OWASP Top 10

Daily commitment: 3 hours/day

- Install Burp Suite Community Edition (free) — the primary tool for web app testing
- Configure your browser to route traffic through Burp's proxy — now you see every HTTP request
- SQL Injection: what it is, how to detect it, basic manual testing, use of SQLMap tool
- Cross-Site Scripting (XSS): stored, reflected, and DOM-based XSS — test for each type
- Insecure Direct Object Reference (IDOR): accessing other users' data by changing an ID in the URL
- Broken Authentication: weak passwords, missing logout, session fixation attacks
- SSRF (Server-Side Request Forgery): make the server fetch resources on your behalf
- PortSwigger Web Security Academy (portswigger.net/web-security): the gold standard free web security course — complete the SQL Injection and XSS labs
- DVWA (Damn Vulnerable Web Application): install locally with Docker, practice attacks safely

Milestone: You can manually test a web application for OWASP Top 10 vulnerabilities using Burp Suite and document findings professionally

Week 7: Exploitation & Metasploit

Daily commitment: 3 hours/day

- Metasploit Framework: the most widely used exploitation framework — searchsploit, msfconsole, use, set, run
- Metasploit concepts: exploits, payloads (Meterpreter), auxiliary modules, post-exploitation modules
- Practice: Exploit a vulnerability in Metasploitable2 using Metasploit — gain a shell, enumerate the system
- Password attacks: Hydra for online brute forcing, Hashcat and John the Ripper for offline hash cracking
- Privilege escalation: linPEAS script — automatically check for privilege escalation vectors on Linux
- Wireshark: capture network traffic, analyze protocols, identify credentials sent in plaintext
- Social engineering basics: phishing email construction, pretexting — understand how human attacks work
- TryHackMe: complete the 'Jr Penetration Tester' learning path — 64 hours of structured content

Milestone: You have successfully exploited a vulnerable machine in a lab environment using Metasploit and escalated privileges

Week 8: CTF Practice & Blue Team Basics

Daily commitment: 3 hours/day

- Capture The Flag competitions: Hack The Box Starting Point machines (4 guided beginner machines — all free)
- Complete 5 easy Hack The Box machines: each teaches a different attack path
- Blue team introduction: SOC analyst skills — log analysis, SIEM concepts, incident response workflow
- Windows Event Logs: Event ID 4624 (logon), 4625 (failed logon), 4720 (new user created) — critical for threat hunting
- SIEM: Try the Splunk free trial or Wazuh (free open-source SIEM) — set up basic alerting rules
- Introduction to threat intelligence: IOCs (Indicators of Compromise), MITRE ATT&CK framework
- Security report writing: document vulnerabilities in the format used by professional penetration testers — executive summary, technical findings, CVSS score, remediation recommendations

Milestone: You have solved 5 CTF machines on Hack The Box and can write a professional penetration testing report

Phase 3: Job Ready & Specialization

Certifications, portfolio, job applications, and bug bounty basics

Week 9: Portfolio & Certifications

Daily commitment: 2 hours/day

- Create a security portfolio on GitHub: document each TryHackMe room and HTB machine you completed — writeups are your portfolio
- Write-ups format: what was the machine, what vulnerabilities you found, what tools you used, what the exploit was, what you learned
- TryHackMe profile: a strong profile with Top 5% rank is visible to recruiters — they DO look at this
- Start CompTIA Security+ exam prep: use Professor Messer's free course and Darril Gibson's book
- Google Cybersecurity Professional Certificate (Coursera): 6-course program, widely recognized, includes tools labs
- Create LinkedIn profile with focus on security: list tools (Kali, Burp Suite, Nmap, Metasploit, Wireshark), TryHackMe profile link, any CTF competitions you have done

Milestone: You have a professional security portfolio, a TryHackMe profile with 100+ rooms completed, and are halfway through Security+ prep

Week 10: Job Applications — SOC Analyst Route

Daily commitment: 3+ hours/day

- Apply to: SOC Analyst (Tier 1), Junior Security Analyst, IT Security Analyst, Information Security Associate, Vulnerability Analyst roles
- Target companies: Wipro, HCL, Infosys (all have large security practices), MSSPs (Managed Security Service Providers), banks with security operations centers
- SOC interview prep: explain the incident response lifecycle, what is a SIEM, how would you investigate a phishing alert, what is the difference between IDS and IPS
- Prepare for: Security+ exam within 3 months — it is often required or preferred for SOC Analyst roles
- Reach out to security professionals on LinkedIn: ask about their day-to-day, what certifications they recommend, which companies have good security teams

Milestone: 50+ applications sent and interviews scheduled for SOC Analyst and Junior Security Analyst positions

Week 11: Bug Bounty Basics

Daily commitment: 2-3 hours/day

- What is bug bounty: companies pay ethical hackers to find vulnerabilities in their products before criminals do
- Sign up for HackerOne and Bugcrowd (both free) — the two biggest bug bounty platforms
- Start with public programs with beginner-friendly scope: targets like Hackerone's public programs for website apps
- Focus on IDOR and XSS bugs first — the most commonly found and rewarded for beginners
- Do NOT test any system without explicit permission — even if you find a bug ethically, you need the program to explicitly allow it
- Bounty amounts in India: informational bugs $50-200, low $200-1000, medium $1000-5000, high $5000-25000, critical $25000+
- Set realistic expectations: your first bug bounty payout may take 3-6 months. Persistence is everything.

Milestone: You have submitted at least one bug bounty report (even if it is a duplicate or informational — the process matters)

Week 12: Specialization & Advanced Certifications

Daily commitment: 2 hours/day

- Pick your specialization track: (1) Blue Team / SOC — focus on SIEM, threat hunting, incident response, (2) Web App Pentesting — focus on PortSwigger labs, bug bounty, advanced Burp Suite, (3) Network Security — focus on firewalls, IDS/IPS, network forensics
- CEH (Certified Ethical Hacker): the most recognized offensive security cert in India — prepares well for pentesting interviews. Exam cost ~INR 30,000-50,000 (training + exam).
- eJPT (eLearnSecurity Junior Penetration Tester): best beginner pentesting cert, ~USD 200. Practical exam — you actually hack a lab environment.
- Join security communities: OWASP India chapter meetings, nullcon conference, c0c0n conference in India
- Apply for internships at security firms: even a 3-month internship creates immense credibility

- Keep practicing: HTB and TryHackMe release new machines weekly — 30 minutes of daily practice compounds fast

Milestone: You have a specialization, a clear certification roadmap, and are actively networking in the Indian security community

Free Resources

Resource Name	Type	Cost	Why Use It
TryHackMe (tryhackme.com)	Learning Platform	Free (limited)	The #1 beginner-friendly security learning platform — browser-based labs, no setup required. Free tier gives access to many rooms.
PortSwigger Web Security Academy	Online Course	Free	The definitive free web application security course — 210+ labs, created by the makers of Burp Suite
Professor Messer CompTIA Security+ Course	YouTube/ Website	Free	The gold standard free Security+ study material — clear, organized, exam-focused
Hack The Box Starting Point	Lab Platform	Free (4 machines)	4 guided beginner machines — structured introduction to real-world hacking
OverTheWire Wargames (overthewire.org)	Platform	Free	Classic hacking challenges organized by difficulty — Bandit is the best Linux starter
IppSec YouTube Channel	YouTube Channel	Free	Detailed walkthroughs of Hack The Box machines — watch before attempting, learn the methodology
OWASP Top 10 (owasp.org)	Documentation	Free	The official list of the 10 most critical web vulnerabilities — required reading for all security professionals
DVWA (Damn Vulnerable Web App)	Lab Software	Free	Run a deliberately vulnerable web app locally — practice SQL injection, XSS, CSRF, and more safely
Cybrary (Free Courses)	Online Courses	Free (limited)	Wide range of security courses — SOC Analyst path and CompTIA Security+ prep are available for free
TCM Security (tcm-sec.com) YouTube	YouTube Channel	Free	Practical ethical hacking content from former professionals — PNPT exam prep content is excellent

Paid Resources (Worth the Money)

Resource Name	Type	Cost	Why Use It
TryHackMe Premium	Platform	~INR 750/month	Full access to all learning paths — the Jr Penetration Tester path alone justifies the cost
TCM Security — Practical Ethical Hacking	Course	~USD 30 (~INR 2500)	The best practical pentesting course for beginners — real lab environment, not just theory
eJPT Certification (INE Security)	Cert	~USD 200 (~INR 16,500)	Best beginner pentesting certification — practical exam where you actually hack a lab. Widely respected.
CompTIA Security+ Exam (SY0-701)	Exam	~USD 392 (~INR 32,000)	The most recognized entry-level security cert — significantly helps for SOC Analyst roles. Study free with Professor Messer.

Projects / Portfolio Items to Build

Project 1: Home Security Lab

- What you build: Set up a VirtualBox environment with 3 VMs: Kali Linux (attacker), Metasploitable2 (vulnerable target), and Windows 10 (optional). Document your setup in GitHub with network diagram.
- Skills demonstrated: VirtualBox, Kali Linux, networking configuration, documentation
- Estimated build time: Weekend project (~5 hours)
- Best done in: Phase 1

Project 2: Network Scan & Vulnerability Report

- What you build: Scan Metasploitable2 using Nmap, identify all open ports and services, research each service for known vulnerabilities (use CVE Details or NVD), and write a formal vulnerability assessment report.
- Skills demonstrated: Nmap, vulnerability research, professional report writing, CVSS scoring
- Estimated build time: 3-5 days
- Best done in: Phase 1-2

Project 3: Web Application Pentest Report

- What you build: Test DVWA (Damn Vulnerable Web Application) for all OWASP Top 10 vulnerabilities using Burp Suite. Document each finding: description, severity, reproduction steps, impact, and remediation recommendation.
- Skills demonstrated: Burp Suite, DVWA, OWASP Top 10, professional pentest reporting
- Estimated build time: 1 week
- Best done in: Phase 2

Project 4: CTF Writeup Collection

- What you build: Solve 10+ TryHackMe rooms and 3+ Hack The Box machines. Write detailed writeups for each: what was the challenge, your methodology, tools used, commands run, lessons learned. Publish on GitHub or a personal blog.
- Skills demonstrated: TryHackMe, Hack The Box, Nmap, Metasploit, Burp Suite, documentation
- Estimated build time: Ongoing throughout course
- Best done in: Phase 1-3

Project 5: Python Security Tool

- What you build: Build a Python tool that performs a basic network scan: given an IP range, discovers live hosts using ping, scans common ports, and outputs a formatted report. Add a feature to check if default credentials work on discovered services.
- Skills demonstrated: Python (socket, subprocess, requests), network programming, security automation
- Estimated build time: 1 week
- Best done in: Phase 3

Career Outcomes

Job Title	Experience	Salary (India)	Primary Skills
SOC Analyst Tier 1	0-1 year	INR 4-8 LPA	SIEM, log analysis, incident response basics
Junior Penetration Tester	0-1 year	INR 5-10 LPA	Web app testing, Burp Suite, Nmap, reporting
IT Security Analyst	1-2 years	INR 7-14 LPA	Vulnerability management, compliance, monitoring
Security Engineer	2-4 years	INR 14-25 LPA	Security architecture, tooling, automation
Penetration Tester	2-5 years	INR 15-30 LPA	Full pentest methodology, report writing, OSCP
Security Consultant	4+ years	INR 25-50 LPA	Client-facing, multiple domains, team lead
CISO (Chief Info Sec Off.)	10+ years	INR 50-1Cr+	Strategy, governance, board-level communication
Bug Bounty Hunter	1+ year	Variable/freelance	Web app hacking, responsible disclosure, persistence

Certifications Roadmap (in order)

- CompTIA Security+ (SY0-701) — First certification. Entry-level, vendor-neutral, globally recognized. Study free with Professor Messer.
- CEH v12 (Certified Ethical Hacker) — Most recognized offensive cert in India for jobs. EC-Council provides training. Exam ~INR 50,000 total.
- eJPT (eLearnSecurity Junior Penetration Tester) — Best practical beginner cert. ~USD 200. You hack a real lab environment.
- OSCP (Offensive Security Certified Professional) — The gold standard for penetration testers. Hard, practical exam. 24-hour hacking challenge. ~USD 1499.
- CISSP — For senior roles and management track. Requires 5 years of experience. Most prestigious security certification globally.

Appendix — Cross-Course Reference

India Salary Comparison — All 5 Courses

Course	Fresher (0-1yr)	Mid (2-4yr)	Senior (5+yr)	Freelance Rate
GenAI & Prompt Engineering	INR 6-12 LPA	INR 12-25 LPA	INR 25-45+ LPA	INR 1500-5000/hr
DevOps & Cloud Engineering	INR 6-14 LPA	INR 15-25 LPA	INR 30-60+ LPA	INR 2000-6000/hr
Frontend Development	INR 3.5-8 LPA	INR 10-20 LPA	INR 22-50 LPA	INR 500-3000/hr
Data Analytics	INR 3.5-7 LPA	INR 8-15 LPA	INR 15-30+ LPA	INR 1000-3000/hr
Cybersecurity & Ethical Hacking	INR 4-8 LPA	INR 10-20 LPA	INR 25-60+ LPA	Variable (bug bounty)

Top Free Learning Platforms Summary

Platform	Best For	URL
TryHackMe	Cybersecurity — beginners	tryhackme.com
Hack The Box	Cybersecurity — intermediate	hackthebox.com
PortSwigger Academy	Web application security	portswigger.net/web-security
The Odin Project	Frontend development	theodinproject.com
freeCodeCamp	Frontend, Data, Python	freecodecamp.org
Kaggle	Data Analytics, Python	kaggle.com
KodeKloud	DevOps, Cloud	kodekloud.com
DeepLearning.AI	GenAI, AI Engineering	deeplearning.ai
roadmap.sh	All courses — visual roadmaps	roadmap.sh
Linux Journey	Linux fundamentals	linuxjourney.com

Recommended YouTube Channels by Course

Course	Channel Name	What to Watch First
GenAI	Andrej Karpathy	'Let's build GPT from scratch' — understand how LLMs work at a deep level
GenAI	freeCodeCamp	'LangChain Full Course' — build your first LangChain app

Course	Channel Name	What to Watch First
DevOps	TechWorld with Nana	'Docker Tutorial for Beginners' — the clearest Docker explanation
DevOps	KodeKloud	'Kubernetes for Beginners' — step-by-step K8s introduction
Frontend	Kevin Powell	'Flexbox in 100 seconds' — start here before anything else in CSS
Frontend	Traversy Media	'JavaScript Crash Course' — best 2-hour JS intro for beginners
Data	Alex the Analyst	'Data Analyst Bootcamp' — full free course covering SQL, Python, Power BI
Data	StatQuest Josh Starmer	'Statistics Fundamentals' playlist — makes stats actually understandable
Cybersecurity	Professor Messer	'CompTIA Security+ 701 Course' — free, complete exam prep
Cybersecurity	IppSec	Any Hack The Box walkthrough video — learn professional methodology

Job Search Strategy — All Courses

Where to Search in India

- LinkedIn — Most important platform. Set job alerts for your target role. 'Easy Apply' for volume, custom applications for target companies.
- Naukri.com — India's largest job portal. Upload an updated resume and apply daily. Also good for salary research.
- Instahyre — Product companies list here. Fewer jobs than Naukri but higher quality for tech roles.
- AngelList India (Wellfound) — Startup jobs. Many early-stage companies post here offering equity and responsibility.
- LinkedIn cold outreach — Connect with engineering managers, team leads, and HR at target companies. A personalized message beats 100 cold applications.

Resume Tips for Tech Roles

- One page maximum for 0-3 years of experience. Two pages only after 5+ years.
- Use keywords from job descriptions — Indian companies often use ATS (Applicant Tracking Systems).
- Lead with projects if you have no corporate experience — employers care about what you have built.
- Quantify everything: 'Built a React app used by 500+ students' beats 'Built a React app'.
- GitHub link on your resume is non-negotiable for development roles.

--- End of Course Catalog Document ---

Version 1.0 | Research sourced from: roadmap.sh, Kaggle, KodeKloud, TryHackMe, DeepLearning.AI, Google Career Certificates, multiple salary surveys (Glassdoor, Naukri, LinkedIn, NASSCOM) | Updated April 2026